

THE OPERATING SYSTEM FOR FINANCE OPERATIONS

One conversation. A live agent pipeline. P2P, O2C and R2R on the same engine.

Nexus is not a single-use 3-way matcher with extra screens bolted on. A user describes any finance process in plain English, uploads working files and optional policy, and Nexus interviews them while the canvas builds — Ingestion, Matching, Rules & Math, Decision and Output, used only when needed, repeated when the process is complex.

Two proofs of the same product: “Match invoices to POs and receipts, flag anything over 2% or \$50” and “Read unread purchase-request mail, judge it against policy, draft a reply I send myself.” Neither is a hardcoded template.

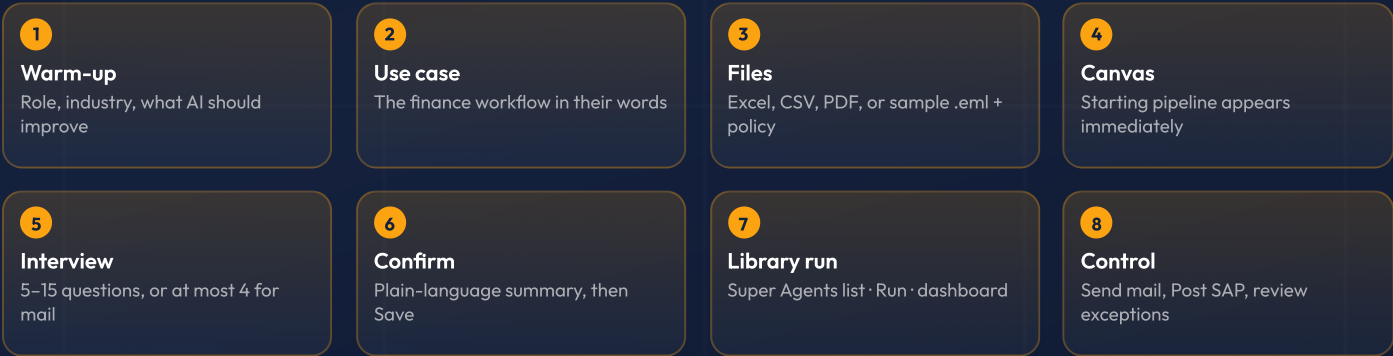
The work Nexus takes off the floor

Re-keying invoices and POs. Matching bank to GL by eye. Recomputing the same tolerances. Chasing approvals over email. Rebuilding the close pack every cycle. When a rule changes, teams hunt every spreadsheet. Nexus stores the rule as versioned behavior — tune it, compare it, roll it back.

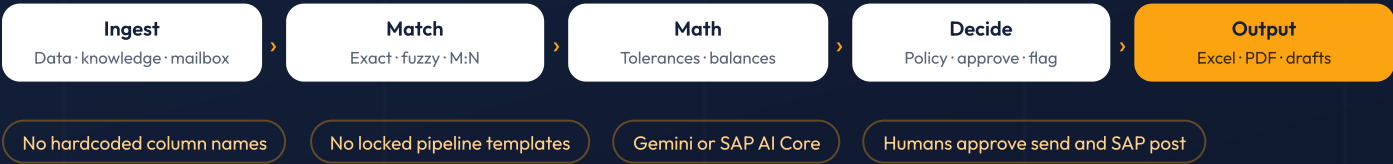
Who sits in the room

- **Ops manager** — every run, every exception
- **Controller / CFO** — governed, cited rules
- **Analyst** — design without writing code
- **Auditor** — replay what ran and why

FROM CHAT TO SUPER AGENT — IN ONE SITTING



FIVE SPECIALISTS — COMPOSE, DO NOT PAD



Complete capability — one agent, many modes

Complexity lives in modes, not in extra products. Nexus picks the mode; the user can change it. Unused agents stay off the canvas. Complex close packs can chain the same agent twice.

IN

Ingestion

The only agent that touches the outside world. Everything downstream consumes its envelope.

- Data mode** — Excel, CSV, PDF records into typed tables
- Knowledge mode** — policies, SOPs, contracts into rules and facts (long PDFs are chunked, not dumped wholesale)
- Mailbox mode** — sample .eml teaches layout; run-time IMAP fetches unread PR/PO mail and skips noise

MA

Matching

Answers “are these the same thing?” Structural when keys are clean; semantic when vendors and dates drift.

- Exact, composite and temporal / windowed joins
- Fuzzy entity normalize (“Dell Inc” vs “Dell Computers”)
- Split and M:N allocation (one receipt, many invoices)
- Directional AR/AP, reversals, distinct-entity guard

MT

Rules & Math

Plain English becomes exact, money-safe logic. The user never writes code; auditors can inspect the compiled rule.

- Per-row, aggregate, sequential (running balance)
- Gates: 2% or \$50, stale items, budget breach
- Precision, rounding, empty-value and fail strategy

DE

Decision

LLM judgment with a cited rationale — anomaly class, policy interpretation, or approval gateway.

- Approved · Flagged · Escalated streams
- Advisory by default: humans still sign off exceptions
- Mailbox: verdict writes the draft reply after the judge, never a placeholder copied into the email

OUT

Output

Multi-tab Excel, visual PDF briefings, and — for mail — dashboard drafts. SMTP and SAP are buttons, not pipeline side-effects.

Q

Smart interview

Minimum 5 questions for spreadsheets (cap 12–15). Mailbox: at most 4, about that mail. Unclear after the cap → expert handoff, no guessing.

TR

Trace & tune

Every run stores agent, behavior version, in/out, duration. Chat and config panel stay one source of truth. LLM: Gemini or SAP AI Core.

Matcher ports on the canvas. Matched groups flow to math or decision. Residuals (short-pays, unapplied cash) flow to netting. Exceptions and unmatched pairs flow to Decision or an exception workbook. The user sees the split; they do not write glue code.

WHAT “UNIVERSAL ENGINE” MEANS IN A SALES CONVERSATION

They may assume	What Nexus actually does
Pre-built 3-way SKU	3-way match is a composed graph: three data ingests + policy knowledge + matcher + math gate + decision + Excel. Same agents build bank recon tomorrow.
Hardcoded PO_NUMBER	Schemas are detected from the file. Match keys come from the interview (“PO number and line ID”), stored as versioned config.
Auto-email / auto-post ERP	Drafts stay draft. Send and Post in SAP are human clicks. Spreadsheet journeys still end in files unless a reviewer acts.
One LLM, one path	Switch Gemini or SAP AI Core. Mailbox still locks Matcher/Math off so Decision is not stranded behind an empty matcher.

Invoice 3-way matcher — the flagship AP flow

An AP manager wants invoices matched to purchase orders and goods receipts, variances flagged, and a pack the controller can sign. Nexus does not ship a fixed “3-way product.” It composes the five agents from the conversation and the files.

Files in: PO_Export.xlsx · Goods_Receipts.csv · Invoices.pdf · AP_Procurement_Policy.pdf | **Ask:** match on PO + line, fuzzy vendors, flag over 2% or \$50, Excel for AP plus a summary.

Ingest PO

Ingest GR

Ingest invoice

Policy

Matcher structural + semantic

Math 2% / \$50

Decision + Excel pack

Matched

Clean 3-way groups to Math, then Decision for policy extras (rush freight, accessories).

Residuals

Quantity or price leftover after a partial receipt — not silently dropped.

Unmatched

Missing GR, duplicate invoice, vendor not on PO — exception tab with reasons.

Interview (typical)	What the canvas learns
Q1 Keys	PO Number + PO Line Item ID across all three documents
Q2 Vendors	Semantic normalize so “Dell Computers” still matches “Dell Inc”
Q3 Variance	Math gate: price or qty variance above 2% or \$50 is a flag
Q4 Policy extras	Decision reads the AP SOP for freight, tax, and non-PO spend
Q5 Output	Tab 1 matched, Tab 2 exceptions with plain-English why, Tab 3 residual audit

Bank reconciliation with running balances

Treasury uploads the bank statement and GL cash extract plus a close SOP. Nexus chains Math: stated vs computed balance, then a stale-item Decision. No 3-way keys — temporal match (date ± 2 days) plus amount.

Bank file

GL cash

Temporal matcher

Running balance math

Stale / break decision

Certified PDF + Excel

Also on this engine

- Cash application — M:N payments to invoices, residuals, reversals
- Intercompany AR/AP — directional who-owes-whom, in-transit breaks
- Budget vs actual — variance math + Decision narrative for FP&A
- If the contract is contradictory past question 15 — graceful expert handoff with the canvas draft saved

What “good” looks like in the demo

- Canvas lights up after upload, not after a 20-question form
- Matcher and Math nodes appear only because the AP story needs them
- Variance rule is editable in the panel and echoed in chat
- Run trace shows which behavior version produced each flag

Purchase requests from email — then human Send and SAP

Same Architect as 3-way match. Working file is a sample .eml plus IT / purchase policy. Matcher and Math stay off unless this conversation asks for them. Extraction is from each live message — not from laptop, city, or cost-centre constants.

.eml + policy

IMAP unread only

Keep PR / PO subjects

Decision vs policy

Draft on dashboard

Send and / or Post SAP

Control	Reviewer does	Nexus never does
Review	Double-click row (inline expander). Confirm OrderQuantity, NetPriceAmount, PurchaseOrderItemText. Read Why this result.	Auto-write a placeholder draft before Decision. Hide the table in a mail popup.
Send / Send all	SMTP the approved draft from the ingest node's mailbox settings.	Email because the pipeline finished.
Don't send	Mark skipped — will not go out.	Send skipped rows later by accident.
Post in SAP	OData A_PurchaseOrder. Popup shows PurchaseOrder from JSON (or d.PurchaseOrder).	Post ERP without a click. Mix Send and Post as one button.

COVERAGE — ONE LIBRARY, THREE DOMAINS

Domain	Workflows Nexus composes	Show this
P2P	3-way match, duplicate invoices, contract / rate checks, PO vs non-PO paths, PR mail → draft → optional S/4 PO, payment exception packs.	Journey 1 matcher · Journey 3 mailbox
O2C	Cash application (M:N), deductions, aging, unapplied cash, billing accuracy, revenue evidence checks.	Allocation matcher + residuals
R2R	Bank vs GL, running balances, intercompany AR/AP, journal policy, budget vs actual, close reporting.	Journey 2 bank recon

WHY THIS IS A DIFFERENT CONVERSATION THAN RPA OR A MATCHER SKU

A conversation, not a 40-field form
Asks only what it still does not know; the graph updates live.

The LLM decides the next step
Enough → build. Missing a rule → ask. Still unclear → expert, no guess.

Agents, not macros
3-way uses Matcher. Mail uses Decision. Bank chains Math. Same product.

Humans keep the keys
Exceptions, SMTP and S/4 posts are explicit. Audit trail on every run.

SPEED
Days of matching and inbox chasing become minutes of review.

ACCURACY
Exact math plus policy cited on every verdict.

CONTROL
Send, Don't send, Post in SAP — never silent side-effects.

SCALE
One agent library. Unlimited processes. Versioned rules.

Run two demos in one sitting.
First: invoice 3-way match — three files, policy, live matcher, 2% / \$50 gate, exception workbook. Second: unread PR mail — four interview questions, dashboard drafts, Send, then Post in SAP and the PurchaseOrder popup. That pair is the product: one Architect, two graphs, humans still in charge.
Start · python run.py → Architect (Create Agent) · Full specification · product-feature.md